ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE E

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET Mechatronic Systems paper (project based, 100 marks): design specifications, automating a punching machine with a gravity magazine, sensors and a timer, fastening, bills of materials and costs, project planning with a Gantt chart and block diagram, testing and commissioning, and maintenance and remote monitoring.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw sketches, charts and circuits neatly. Use only a black or blue pen.

> **PROJECT.** A workshop punches two holes in small steel plates with a manually operated punching machine. An operator places each plate by hand and presses a button. The manager wants the machine automated: a gravity magazine feeds the plates, a pneumatic cylinder pushes each plate into position, a sensor confirms that a plate is in place, the punch operates for a set time, and finished plates drop into a bin.

QUESTION 1

1.1 Explain the purpose of writing a design specification before a project starts. (2)

1.2 Name FOUR headings (aspects) that should appear in the design specification of the automated punching machine, and write ONE question that must be answered under each heading. (8)

[10]

QUESTION 2

2.1 Make a neat freehand sketch of the automated punching machine. Show the gravity magazine, the pusher cylinder, the plate-present sensor, the punch and the output chute. (5)

2.2 The machine is now switched on and off by the latching circuit below. Redraw and modify it so that the punch (solenoid Y1) only operates when a plate is detected by sensor S1, and stays on for 3 seconds controlled by timer T1. (6)

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POS ─┬──[ Pb1 START (NO) ]──┬──[ Pb2 STOP (NC) ]──( CR )── NEG
     └──[ CR 1.1 (NO) ]─────┘
POS ────[ CR 1.2 (NO) ]──────────────────────────( M )─── NEG
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2.3 Explain the difference between permanent and temporary (removable) fastening methods, and give TWO examples of each. (4)

2.4 List FOUR items of information that must appear in a bill of materials (BOM). (4)

2.5 Explain the difference between the capital cost and the running cost of the project. (2)

2.6 Draw the symbol of a 3-wire PNP inductive proximity sensor. (2)

2.7 Explain how the gravity magazine increases the productivity of the machine. (2)

[25]

QUESTION 3

3.1 List FIVE skills or tools that a technician needs to assemble this machine. (5)

3.2 Draw up a Gantt chart for completing the project in three weeks under these headings: PLANNING, DESIGN, PROCUREMENT, ASSEMBLY, WIRING AND PROGRAMMING, TESTING AND COMMISSIONING. (10)

3.3 Explain the purpose of a risk assessment before the project starts, and list THREE risks in this project. (5)

3.4 Explain the difference between a circuit (schematic) diagram and a wiring diagram. (4)

3.5 Draw a system block diagram of the automated punching machine, showing its inputs, controller and outputs. (6)

[30]

QUESTION 4

4.1 Explain the purpose of a dry run (running the machine without plates) before production starts. (2)

4.2 List THREE checks that must be done before power and air are switched on for the first time. (3)

4.3 Explain what is meant by commissioning. (2)

4.4 Explain why the position of the plate-present sensor may need adjusting during commissioning. (2)

4.5 Use a spider diagram to show SIX areas that must be tested on the automated punching machine. (6)

[15]

QUESTION 5

5.1 Draw up a weekly inspection checklist for the automated punching machine with at least SIX items. (6)

5.2 Explain how the outputs of the PLC can be tested safely during scheduled maintenance. (3)

5.3 List FOUR things that the operator's manual of the machine must contain. (4)

5.4 Explain what a short circuit is. (1)

5.5 Explain TWO ways in which a remote monitoring system can alert a technician to a fault, and give ONE advantage of each. (4)

5.6 Explain why faults should be logged (recorded). (2)

[20]

TOTAL: 100

END OF QUESTION PAPER